

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20 Total Marks:20 Annexure A

Mock Interview

Code of Conduct:

I CH.V.Ramayogi(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:CH.V.Ramayogi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	

Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	
-----------------------	----	--	---	------------------------------------	---------------------------------	--

Annexure A

Mock Interview

1. What is Eigenfaces?
2. Define PCA in image processing.
3. What is contour-based shape representation?
4. What is region-based shape representation?
5. How does an active contour (snake) work?
6. How does a level set method work?
7. What is the Hough Transform?
8. How are Fourier descriptors used in shape analysis?
9. How are wavelet descriptors used in shape analysis?
10. What is a particle filter?
11. How is foreground–background separation done in surveillance systems?
12. How do multi-camera systems track pedestrians?
13. What is a Convolutional Neural Network (CNN)?
14. How does a Vision Transformer (ViT) work?
15. What is a Generative Adversarial Network (GAN)?
16. How is a Support Vector Machine (SVM) used for image classification?
17. How are Hidden Markov Models (HMM) applied in activity recognition?
18. What is a Gaussian Mixture Model (GMM)?
19. How is graph-based learning applied to 3D shape correspondence?
20. How is road and lane detection performed in autonomous driving?

ANSWERS:

1. What is Eigenfaces?

Answer: Eigenfaces is a face-recognition technique based on PCA (Principal Component Analysis). It converts face images into a smaller set of important features called eigenfaces and compares them for recognition.

2. Define PCA in image processing.

Answer: PCA is a dimensionality-reduction technique. It finds the most important variations in an image and represents the image using fewer features while retaining most of the useful information.

3. What is contour-based shape representation?

Answer: Contour-based representation describes an object using its boundary or outline. Examples include chain codes, Fourier descriptors, and polygonal approximation.

4. What is region-based shape representation?

Answer: Region-based representation describes the entire area inside an object, not just its boundary. Features such as area, centroid, moments, and texture can be used.

5. How does an active contour (snake) work?

Answer: An active contour is a curve placed around an object. It moves toward the object's boundary by minimizing an energy function based on internal smoothness and external image information.

6. How does a level set method work?

Answer: A level set represents a contour as a mathematical function. The contour can automatically expand, shrink, split, or merge depending on the image features.

7. What is the Hough Transform?

Answer: Hough Transform is a feature-detection technique used to detect shapes such as lines, circles, and curves. It converts image points into a parameter space and finds shapes by looking for maximum votes.

8. How are Fourier descriptors used in shape analysis?

Answer: Fourier descriptors represent an object's boundary using Fourier coefficients. They are useful for comparing shapes because the coefficients capture the overall shape information.

9. How are wavelet descriptors used in shape analysis?

Answer: Wavelet descriptors analyze a shape at different scales or resolutions. They can capture both global and local shape details.

10. What is a particle filter?

Answer: A particle filter is a probabilistic tracking algorithm. It uses many possible states, called particles, to estimate the position of a moving object even when there is noise or uncertainty.

11. How is foreground-background separation done in surveillance systems?

Answer: The system separates moving objects from the background using techniques such as background subtraction, frame differencing, or Gaussian Mixture Models. The detected foreground can then be used for object tracking.

12. How do multi-camera systems track pedestrians?

Answer: Multiple cameras detect and track the same pedestrian from different views. They use person detection, appearance features, camera calibration, and identity matching to maintain the person's identity across cameras.

13. What is a Convolutional Neural Network (CNN)?

Answer: CNN is a deep-learning network mainly used for image processing. Convolution layers extract features, pooling reduces dimensions, and fully connected layers perform classification.

14. How does a Vision Transformer (ViT) work?

Answer: ViT divides an image into small patches, converts them into embeddings, and processes them using a Transformer with self-attention. This allows the model to learn relationships between different image regions.

15. What is a Generative Adversarial Network (GAN)?

Answer: GAN consists of two neural networks: a Generator and a Discriminator. The generator creates fake images, while the discriminator tries to distinguish real images from fake ones. Both improve through competition.

16. How is an SVM used for image classification?

Answer: SVM classifies images by finding the best separating hyperplane between different classes. Image features are given as input to the SVM, which predicts the class of a new image.

17. How are HMMs applied in activity recognition?

Answer: HMM represents activities as a sequence of hidden states. For example, walking can be represented as a sequence of body movements. The model uses observed features to determine the most likely activity.

18. What is a Gaussian Mixture Model (GMM)?

Answer: GMM is a probabilistic model that represents data using a mixture of multiple Gaussian distributions. In computer vision, it is commonly used for background modeling and foreground detection.

19. How is graph-based learning applied to 3D shape correspondence?

Answer: A 3D shape can be represented as a graph, where vertices represent points and edges represent relationships between them. Graph-based learning finds corresponding points between different 3D shapes based on their structural and geometric features.

20. How is road and lane detection performed in autonomous driving?

Answer: Cameras capture road images, and image-processing or deep-learning techniques detect road boundaries and lane markings. Methods such as edge detection, Hough Transform, CNNs, and segmentation can be used to identify lanes and help the vehicle stay on the correct path.